

Final Exam

Diya Jain

Ques 2. $G_i(k) = k_i$ where $0 \leq i < 9$, $K = \{0, 1\}$

$$G_i(k) = (k_{i \bmod 10} \oplus (i \bmod 2)) \oplus G_{i-1}(k) \text{ for } i \geq 10$$

$$k = 1100011111$$

$$1. G_{0-9}(k) = \overset{0}{1} \overset{1}{1} \overset{2}{0} \overset{3}{0} \overset{4}{0} \overset{5}{0} \overset{6}{1} \overset{7}{1} \overset{8}{1} \overset{9}{1}$$

$$G_{10}(k) = (k_{10 \bmod 10} \oplus (10 \bmod 2)) \oplus G_9(k)$$

$$= (1 \oplus 0 \oplus 1) = 0$$

$$G_{11}(k) = (k_{11 \bmod 10} \oplus (11 \bmod 2)) \oplus G_{10}(k)$$

$$= (1 \oplus 1 \oplus 0) = 0$$

$$G_{12}(k) = (k_{12 \bmod 10} \oplus (12 \bmod 2)) \oplus G_{11}(k)$$

$$= (0 \oplus 0 \oplus 0) = 0$$

$$G_{13}(k) = (k_{13 \bmod 10} \oplus (13 \bmod 2)) \oplus G_{12}(k)$$

$$= (0 \oplus 1 \oplus 0) = 1$$

$$G_{14}(k) = (k_{14 \bmod 10} \oplus (14 \bmod 2)) \oplus G_{13}(k)$$

$$= (0 \oplus 0 \oplus 1) = 1$$

$$G_{15}(k) = (k_{15 \bmod 10} \oplus (15 \bmod 2)) \oplus G_{14}(k)$$

$$= (1 \oplus 1 \oplus 1) = 1$$

$$G_{16}(k) = (k_{16 \bmod 10} \oplus (16 \bmod 2)) \oplus G_{15}(k) \\ = (1 \oplus 0 \oplus 1) = 0$$

$$G_{17}(k) = (k_{17 \bmod 10} \oplus (17 \bmod 2)) \oplus G_{16}(k) \\ = (1 \oplus 1 \oplus 0) = 0$$

$$G_{18}(k) = (k_{18 \bmod 10} \oplus (18 \bmod 2)) \oplus G_{17}(k) \\ = (1 \oplus 0 \oplus 0) = 1$$

$$G_{19}(k) = (k_{19 \bmod 10} \oplus (19 \bmod 2)) \oplus G_{18}(k) \\ = (1 \oplus 1 \oplus 1) = 1$$

$$G(k)_{0, \dots, 19} = 11000111110001110011$$

2. $m = 00111001101001010000$

$$G(k) = 11000111110001110011$$

$$\text{Encrypt} = m \oplus G(k) \\ = 1111110011000100011$$

3. The adversary A predicts $G(k)_{10}$ using the formula $G(k)_1 \oplus G(k)_{10}$.

$$\text{Prediction: } G(k)_1 \oplus G(k)_{10} = 1 \oplus 0 = 1$$

From actual calculation, $G(k)_{10} = 0$, the adversary has a 0% probability of being correct.

$$P(A \text{ outputs } G(k)_{10}) = 0$$

4. Advantage is $\left| \Pr(\text{success}) - \frac{1}{2} \right| = \frac{1}{2}$

Ques 3. $X = \{0, 1, 2, 3, 4, 5\}$

1. There are $6! = 720$ permutations
2. There are 6^6 possible function $X \rightarrow X = 46656$ functions
3. There are $6 \times 5 \times 4 \times 6 \times 6 \times 6 = 25920$ possible functions such that $f(0), f(1), f(2)$ are all different.

4. $P(W_0)$

The number of such functions such that $f(0), f(1)$ and $f(2)$ are all different is 25920.

$$P(W_0) = \frac{25920}{46656} = \frac{5}{9}$$

$P(W_1)$

The challenger draws a key k at random and sends $g_k(0), g_k(1), g_k(2)$. As g_k is a permutation, $g_k(0) \neq g_k(1) \neq g_k(2)$ and the adversary always returns 0.

$$P(W_1) = 1$$

$$\text{Advantage} = \left| \Pr(W_0) - \Pr(W_1) \right| = \left| \frac{5}{9} - 1 \right| = \frac{4}{9}$$

Ques 4. Let $N = 2^{256}$ be the size of the tag space, and let k be the number of messages we draw at random. Let P be the probability that all these messages have a different tag. We have:

$$P = \left(1 - \frac{1}{N}\right) \left(1 - \frac{2}{N}\right) \dots \left(1 - \frac{k-1}{N}\right)$$

As $1-x \leq e^{-x}$, the probability satisfies

$$P \leq e^{-1/N} \cdot e^{-2/N} \cdot \dots \cdot e^{-k-1/N}$$

$$= e^{-\frac{1}{N} \sum_{i=1}^{k-1} i} = e^{-\frac{1}{N} \frac{k(k-1)}{2}}$$

$\therefore$ The formula is $\sqrt{\frac{2 \ln \left(\frac{1}{1-0.05} \right)}{1-0.05}} \cdot \sqrt{N} + 1$

$$= \sqrt{\frac{2 \ln \left(\frac{1}{0.95} \right)}{0.95}} \cdot \sqrt{2^{256}} + 1$$

$$= \sqrt{\frac{2 \ln \left(\frac{1}{0.95} \right)}{0.95}} \cdot 2^{128} + 1$$

$$= 1.089895199 \times 10^{38} + 1$$

$$\approx 1.0898 \times 10^{38} \text{ ans}$$

Ques 5. $N = 490717$, $e = 419383$

$$\frac{e}{N} = \frac{419383}{490717} = 0.85463 = 0 + E_0, \text{ convergent, } \frac{A}{B} = 0$$

$$\frac{1}{E_0} = \frac{1}{0.85463} = 1.170092 = 1 + E_1, \text{ convergent, } \frac{A}{B} = 0 + 1 = 1$$

$$C = \frac{eB - 1}{A} = \frac{419383 - 1}{1} = 419382$$

$$P(x) = x^2 - (N - C + 1)x + N$$

$$= x^2 - (490717 - 419382 + 1)x + 490717$$

$$= x^2 - 71336x + 490717, \text{ no factors.}$$

$$\frac{1}{E_1} = \frac{1}{0.170092} = 5.879171 = 5 + E_2 \quad \text{convergent} = \frac{A}{B} = \frac{0+1}{1+1} = \frac{5}{6}$$

$$C = \frac{eB-1}{A} = \frac{419383(6)-1}{5} = 503259.4$$

$$\begin{aligned} P(x) &= x^2 - (N-C+1)x + N \\ &= x^2 - (490717 - 503259.4 + 1)x + 490717 \\ &= x^2 + 12541.4x + 490717, \text{ no factors} \end{aligned}$$

$$\frac{1}{E_2} = \frac{1}{0.879171} = 1.137435 = 1 + E_3 \quad \text{convergent} = \frac{A}{B} = \frac{0+1}{1+1} = \frac{6}{7}$$

$$C = \frac{eB-1}{A} = \frac{419383(7)-1}{6} = 489280$$

$$\begin{aligned} P(x) &= x^2 - (N-C+1)x + N \\ &= x^2 - (490717 - 489280 + 1)x + 490717 \\ &= x^2 - 1438x + 490717 \end{aligned}$$

$$\frac{-(-b) \pm \sqrt{b^2 - 4ac}}{2a} = \frac{1438 \pm \sqrt{2067844 - 4(1)(490717)}}{2}$$

$$= \frac{1438 \pm \sqrt{104976}}{2} = \frac{1438 \pm 324}{2}$$

$$\frac{1438+324}{2}$$

$$= 881$$

$$\frac{1438-324}{2}$$

$$= 557$$

$$\begin{aligned} p &= 881 \\ q &= 557 \\ d &= 7 \end{aligned}$$

Ques 6. $\text{buff} = [10x^2 + x + 2]$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ m_2 & m_1 & m_0 \end{array}$$

10 is closer to $q_1/2$ than 0, $\therefore m_2$ will be 1.
1 and 2 are closer to 0, $\therefore m_1$ and m_0 will be 0.

The message will be x^2

Ques 7. $e = \begin{bmatrix} 0 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, S = \begin{bmatrix} 27 \\ 87 \\ 99 \end{bmatrix}$

Ques 9. $\begin{pmatrix} 2 & -11 \\ -16 & 90 \end{pmatrix} = A, A^T = \begin{pmatrix} 2 & -16 \\ -11 & 90 \end{pmatrix}$

$$\begin{aligned} A^T \cdot A &= \begin{bmatrix} 2 & -16 \\ -11 & 90 \end{bmatrix} \begin{bmatrix} 2 & -11 \\ -16 & 90 \end{bmatrix} = \begin{bmatrix} (2)(2) + (-16)(-16) & 2(-11) + (-16)(90) \\ (-11)(2) + (90)(-16) & (-11)(-11) + (90)(90) \end{bmatrix} \\ &= \begin{bmatrix} 260 & -1462 \\ -1462 & 8221 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \det(A^T \cdot A) &= (260)(8221) - (-1462)(-1462) \\ &= 16 \end{aligned}$$

$$\text{Vol}(Z) = \sqrt{\det(A^T \cdot A)} = \sqrt{16} = 4$$

$$\begin{pmatrix} 2 & -11 \\ -16 & 40 \end{pmatrix} \xrightarrow{C_2 \rightarrow C_2 + 5C_1} \begin{pmatrix} 2 & -1 \\ -16 & 10 \end{pmatrix} \xrightarrow{C_1 \rightarrow 2C_2 + C_1} \begin{pmatrix} 0 & -1 \\ 4 & 10 \end{pmatrix} \xrightarrow{C_2 \rightarrow C_2 - 2C_1} \begin{pmatrix} 0 & -1 \\ 4 & 2 \end{pmatrix}$$

$$\lambda_1(z) = \sqrt{4^2} = 4$$

$$\lambda_2(z) = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

Ques 1.

1. This is a block cipher
2. $m = 1010 \ 1111 \ 0101$, $k_1 = k_2 = 1$, $k_3 = 1110$, $k_4 = 0111$

$$m_1 = m_1 \oplus (k_1 + k_2)_2 = 1010 + 1 = 1011$$

$$c_1 = m_1 \oplus k_3 = 1011 \oplus 1110 = 0101$$

$$c_2 = m_1 \oplus m_2 \oplus m_3 = 1011 \oplus 1111 \oplus 0101 = 0001$$

$$c_3 = m_3 \oplus k_4 = 0101 \oplus 0111 = 0010$$

$$c = 0101 \ 0001 \ 0010$$

3. $C = 0010 \ 1001 \ 1101$, $k_1 = 0$, $k_2 = 1$, $k_3 = 1110$, $k_4 = 0111$

$$m_1 = c_1 \oplus k_3 = 0010 \oplus 1110 = 1100$$

$$m_1 \oplus (k_1 + k_2)_2 = 1100 \oplus 0 = 1100 = m_1$$

$$m_3 = c_3 \oplus k_4 = 1101 \oplus 0111 = 1010$$

$$m_2 = 1100 \oplus 1001 \oplus 1010 \\ = 1111$$

$$m = 1100 \ 1111 \ 1010$$

4. When $k_1 = k_2 = 1$, the middle block is $c_2 = 0 \dots 0$, in all other cases $c_2 = 1 \dots 1$.

5. When $k_1 = k_2 = 1$, the middle block is $c_2 = 1 \dots 1$, in all other cases $c_2 = 0 \dots 0$.

$$6. P(w_0) = \frac{1}{4}$$

$$P(w_1) = \frac{3}{4}$$

$$\text{Advantage} = \left| \frac{1}{4} - \frac{3}{4} \right| = \frac{1}{2}$$

7. When $n > 2$, $3n > 2n + 2$, and thus the size of the key is less than the size of the message. Therefore, the scheme does not have perfect secrecy.

Ques 8.

PLO 3. Generate MAC: Compute $t = S(m, k)$ using a secret key k for a file m . This tag t is stored or sent with m .

Verify MAC: When m is accessed or received, compute $t' = S(m', k)$ and compare it to the stored t .

Detect Corruption: if $m' \neq m$, then $t' \neq t$, signaling that m was tampered with, likely due to malware.

PLO5. Game 1 ensures authenticity of the message. Game 1 tests whether an adversary can forge a valid tag t for a new message $m \notin \{m_0, \dots, m_e\}$. The adversary's success in this game directly violates the authenticity of the MAC.

Game 2 ensures confidentiality of the message. Game 2 tests whether an adversary can distinguish between two messages m_0 and m_1 , based on their tags t . A success implies that MAC leaks information about the message, violating confidentiality.

PLO4. Game 1 mirrors the real-world scenario where a file m is tampered with by malware to produce m' . If the malware cannot forge a valid tag t' , verification will fail ($V(m', k, t') \neq \text{true}$), and the corruption will be detected.

PLO1. Probability of Winning Game 1.

- The adversary guesses both m and t . Given $S(m, k) = \text{SHA256}(k \parallel \text{SHA384}(k \parallel m))$, the output t is uniformly distributed in 2^{256} (256-bit output from SHA256).
- Probability of a correct guess

$$P = \frac{1}{2^{256}}$$

Number of security bits: $\log_2(2^{256}) = 256$ bits